SARVARBEK ERNIYAZOV

Ph.D. Candidate, Computer Engineering
Reliable Machine Learning: LLM Post-Training and Evaluation; Spatio-Temporal Forecasting
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Academic Profile

I am a Ph.D. candidate in Computer Engineering (dissertation defense 1 December 2026; degree conferral February 2027; available from March 2027). I work on reliable machine learning: the evaluation and post-training of language models, and spatio-temporal forecasting for energy systems. My record includes three first-author AI journal articles (2024–2026) on solar forecasting and industrial-control anomaly detection.

Research Focus

Post-training and evaluation of language models • Tool-calling agents for energy operations • Pre-registered measurement, negative controls and leakage audits • Spatio-temporal and multimodal forecasting • Anomaly detection and distribution shift.

Current Research

- **LLM post-training and evaluation study** (in progress, unpublished; lead researcher). Question: how successive post-training stages change whether a language model adopts dated rule updates supplied in context. Methods: supervised fine-tuning, DPO, reward-model PPO and GRPO with verifier-based rewards; a pre-registered analysis plan; a human-verified, dated rule base; multi-stage training on an H200 with an autonomous agent orchestrating the jobs. Target venues: NeurIPS 2027 Evaluations & Datasets or ACL/EMNLP 2027. Data, results and code are withheld until release.
- **VPP-Copilot** (study in progress): on-device small-language-model tool-calling agent for virtual power plant operations within TIPA RS-2026-25529976. Role: Lead Researcher for the assigned university AI work package; not the formal grant PI.
- **Chinor Bank Assistant**: public pilot (closed; fictional domain). SFT of Qwen3 base models (0.6B–14B) with LoRA, rsLoRA, 4-bit QLoRA and full fine-tuning, then DPO and SimPO on 348 rule-labelled on-policy pairs; frozen 220-item evaluation and 60-attack red-team set; paired bootstrap, McNemar, Holm. SFT accounts for the gain (14B: 45 to 145.0 $\pm$ 2.5 of 200, 5 seeds). DPO and SimPO: no significant difference from the SFT parent detected; the SFT model already passed 86% of its own samples. One annotator; scores measure reproduction of one handbook under a lenient rubric. Code and results: `https://github.com/Sarvarbek-Erniyazov/chinor-bank-assistant-public`

Technical Skills

SFT, LoRA/QLoRA, DPO, reward modeling, PPO, GRPO, verifier design, from-scratch pretraining of small transformers (tokenizers, training loop; FaultLine), pre-registration, evaluation design, multi-stage training on a single H200; spatio-temporal graph learning; multimodal fusion; mixture-of-experts; federated and personalized learning; time-series forecasting; anomaly and drift detection; conformal prediction; uncertainty quantification; selective prediction; computer vision; frequency-domain learning; and edge-aware evaluation.

Education and Availability

Ph.D. in Computer Engineering, Chonnam National University, Yeosu, South Korea 2020 – Feb. 2027 (exp.)
Advisor: Prof. Chang Gyoon Lim • Dissertation defense: **1 December 2026** • Official degree conferral: **February 2027**
Dissertation (working title): *Shift-Resilient Learning for Heterogeneous Energy Systems: From Spatio-Temporal Forecasting to Risk-Calibrated Event Prediction.* Four studies: multi-station solar forecasting (Energy Reports, 2025), multimodal solar forecasting (KSII TIIS, 2026), federated building-energy forecasting (manuscript under review, revised and resubmitted), and wind-turbine event risk (FaultLine). Doctoral coursework GPA: 4.21/4.5.

M.S. in Computer Engineering, Kumoh National Institute of Technology, Gumi, South Korea 2016 – 2018
Information Security Laboratory (QCA Research Group)

B.S. in Computer Engineering (Information Security), Tashkent University of Information Technologies, Uzbekistan 2011 – 2015

My background combines undergraduate information-security training, master's research on low-power QCA circuits within an information-security laboratory, and doctoral work on industrial-control anomaly detection.

Postdoctoral availability: from March 2027, and compatible with later start dates. My university affiliation and grant responsibilities conclude upon degree conferral in February 2027; the funded project continues to 2028, but my personal participation ends at conferral.

Selected Research Impact

- **3 peer-reviewed AI journal articles** (2024–2026), all first-author; **9 peer-reviewed publications** in total with verified records (7 journal, 2 proceedings); one manuscript under review, listed separately.
- **12 public research repositories** (counted 11 November 2026) and **7 hosted research interfaces**; technical contributions to funded Korean national R&D in virtual power plants and edge–cloud energy management.

Research Experience

Ph.D. Researcher — AI Laboratory, Chonnam National University 2020 – present
Department of Computer Engineering, Yeosu, South Korea • Advisor: Prof. Chang Gyoon Lim
Research program: spatio-temporal, multimodal, and distributed learning for energy forecasting and cyber-physical anomaly detection.

- Led first-author research on graph-enhanced temporal and frequency modeling for multi-station solar forecasting, including architecture design, implementation, baseline comparison, ablation analysis, and manuscript preparation.
- Developed dynamic graph and mixture-of-experts methods for integrating heterogeneous solar and meteorological modalities across forecast horizons.
- Investigated federated spatio-temporal mixture-of-experts for heterogeneous building-energy clients; the manuscript is under review and is excluded from the peer-reviewed publication count.
- Developed and evaluated autoencoder-based time-series anomaly detection on the HAI industrial-control-system benchmark.
- Built reproducible research pipelines: data validation, leakage-aware splitting, training, statistical evaluation, ablations, and research-prototype integration.
- Contributed AI methodology and technical deliverables to funded Korean national R&D on virtual power plants and edge–cloud energy management.

Individual contribution statement: For first-author doctoral studies, I led method development, software implementation, experimental design, baseline and ablation evaluation, result visualization, and manuscript preparation. Formal grant governance and institutional PI responsibilities remained with the appointed faculty principal investigator.

Graduate Researcher — Information Security Laboratory / QCA Research Group 2016 – 2018
Kumoh National Institute of Technology, Gumi, South Korea

- Designed and simulated low-power quantum-dot cellular automata arithmetic and reversible circuits; reported more than 50% area reduction for the proposed carry-look-ahead architecture relative to the specified comparison design.

National R&D Technical Leadership

Role on the active project: Lead Researcher for the assigned university AI work package; not the formal grant PI. I led the assigned AI work package from problem formulation and data-interface design through model architecture, leakage-aware validation, and research-prototype integration, coordinating technical interfaces with the faculty PI, domain partners, and software contributors while grant governance remained with the appointed PI.

TIPA RS-2026-25529976 — sLM–LMM Cooperative Virtual Power Plant Platform

[Funded and active | 2026–2028] Industry–academia commercialization R&D for on-device AI and cloud-integrated energy management. **Technical role:** Lead Researcher for the assigned university AI work package; not the formal grant PI.

- Define model and evaluation architecture for cooperation between on-device small language models and cloud-based multimodal models in VPP workflows.
- Lead AI-side interface design connecting telemetry, forecasts, operator tools, and edge–cloud inference components.
- Direct technical validation for assigned machine-learning deliverables and coordinate research-prototype integration with faculty and industry partners.

TIPA RS-2025-02312851 — Preliminary Study for an On-Device AI and Cloud-Integrated VPP Platform

[Funded and completed | 2025] **Technical role:** Lead AI Researcher for the machine-learning feasibility work; not the formal grant PI. Led AI feasibility analysis, model architecture planning, and prototype evaluation supporting the subsequent commercialization-stage project, and translated research requirements into a technically scoped edge–cloud VPP development plan.
Earlier funded programs (research contributor): NRF-2017R1D1A3B03034346; 2021RIS-002; RS-2023-00278623.

Unpublished Studies — Not Publications

No peer-reviewed output is claimed for any of these. FaultLine is a sole-author study on personal hardware and public data and the final study of my dissertation. The sensing-privacy study involves external collaborators, who will be named once public attribution is agreed.

- **FaultLine** (programme closed 20 September 2026): sole-author, pre-registered study: a small transformer pretrained from scratch on wind-turbine telemetry and text for event risk. The one supported hypothesis became inconclusive under a label variant, and a time-since-last-fault baseline beats the model (both exploratory, after closure); the study reports this as its finding. Repository
- **Sensing-privacy measurement study**: empirical evaluation of a privacy claim attached to a sensing modality. Manuscript in preparation for double-blind submission; title, results and code are withheld to preserve anonymity.

Selected Open Research Artifacts

Public methodological studies, reproducibility instruments, and research prototypes; none is a peer-reviewed publication or an operational deployment. Index: sarvarbek-erniyazov.github.io.

DriftSentinel — Self-Auditing Distribution-Shift Monitoring

Executable comparison of PSI, KS and chi-square tests with BH-FDR correction, CUSUM, MMD, and BBSD on a 101,766-encounter clinical model. Uses 20-seed negative controls to identify false drift alarms and invalid detector behavior; overturns an initial "8/8 critical" result. Includes 93 automated tests, continuous integration, and determinism checks. Code • Hosted monitoring console

- **ReproLab** — AI4I-2020 reproducibility audit; F1 ceiling 0.9558 derived three ways, spread 0.0015. Code
- **OCRipple** — OCR-error propagation study; ANLS 0.848, layout mAP@50 73.8%. Code

Also: CellTriage, WaveSeg, FreqFD, DALM-DETR, FinRiskGuard, Auto Damage Segmentation, Uzbek Handwriting OCR.

Publications

Verified publication records are listed below, with manuscripts under review reported separately and excluded from peer-reviewed publication counts. The citation count for [J4] was confirmed on Google Scholar on 11 September 2026. My name is shown in bold. Full record: Google Scholar and ORCID.

Record summary 7 peer-reviewed journal articles • 2 peer-reviewed proceedings papers • 1 manuscript under review • first author on every entry

Peer-Reviewed Journal Articles — Published or Accepted

[J1] **S. Erniyazov**, M. A. Jaleel, C. G. Lim, and S. B. Ha. "A multimodal fusion framework for solar forecasting using dynamic GNNs and mixture of experts." *KSII Transactions on Internet and Information Systems*, 20(3), 1337–1360, 2026. doi:10.3837/tiis.2026.03.012
Contribution: first-author multimodal architecture, dynamic graph and expert-routing methodology, evaluation, analysis, and manuscript preparation.

[J2] **S. Erniyazov** and C. G. Lim. "GNN-enhanced temporal patch segmentation and frequency fusion model for robust solar energy production forecasting." *Energy Reports*, 13, 4962–4984, 2025. doi:10.1016/j.egyr.2025.03.063
Contribution: first-author method development, implementation, experimental design, baseline and ablation evaluation, and manuscript preparation.

[J3] **S. Erniyazov**, Y.-M. Kim, M. A. Jaleel, and C. G. Lim. "Comprehensive analysis and improved techniques for anomaly detection in time series data with autoencoder models." *International Journal on Advanced Science, Engineering and Information Technology*, 14(6), 1861–1867, 2024. doi:10.18517/ijaseit.14.6.20451
Contribution: first-author model design, HAI benchmark evaluation, analysis, and manuscript preparation.

[J4] **S. Erniyazov** and J.-C. Jeon. "Carry save adder and carry look ahead adder using inverter chain based coplanar QCA full adder for low energy dissipation." *Microelectronic Engineering*, 211, 37–43, 2019. doi:10.1016/j.mee.2019.03.015
93 citations (Google Scholar, 11 September 2026). Contribution: circuit design, simulation, comparative evaluation, and manuscript preparation.

[J5] **S. Erniyazov** and J. C. Jeon. "Coplanar subtractor design in QCA for arithmetic circuit design." *Advanced Science Letters*, 23(10), 10112–10117, 2017. doi:10.1166/asl.2017.10399

[J6] **S. Erniyazov** and J. C. Jeon. "Implementation of area efficient Fredkin gate for compact reversible QCA circuit." *Advanced Science Letters*, 23(10), 10092–10096, 2017. doi:10.1166/asl.2017.10395

[J7] **S. Erniyazov** and J. C. Jeon. "Reversible circuit design in QCA based on double Feynman gate." *Advanced Science Letters*, 23(10), 9852–9856, 2017. doi:10.1166/asl.2017.9810

Peer-Reviewed Conference or Proceedings Papers

[C1] **S. Erniyazov** and J.-C. Jeon. "Area efficient magnitude comparator based on QCA." *Advanced Science and Technology Letters*, 150, 75–79, 2018. doi:10.14257/astl.2018.150.19

[C2] **S. Erniyazov** and J.-C. Jeon. "Half subtractor in QCA for arithmetic circuits." In *Proceedings of the International Workshop on Future Technology (FUTECH 2017)*, Korean Institute of Information Technology, pp. 148–149, July 2017. Proceedings record

Manuscripts Under Review — Not Included in Publication Count

[M1] **S. Erniyazov** et al. "Fed-ST-MoE: A federated spatio-temporal GNN with mixture-of-experts for heterogeneous building energy and demand forecasting." Manuscript under review (revised and resubmitted).
Investigates decentralized forecasting across heterogeneous buildings with spatio-temporal graph learning and client-aware mixture-of-experts. The study addresses statistical heterogeneity and local-data retention; it does not claim a formal privacy guarantee. Venue and preprint link will be disclosed as permitted by the review policy.

References

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